OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T5:
RetBot, distance control

2026-06-26

Gruppe 38
Marius Off
Janis Graf

Inhaltsverzeichnis

1. Preparations **3**
1.1. (aV) P controller in the time domain 3
1.2. (bV) Transfer function of open loop P controller 3
1.3. (cV) Transfer function of distance control with P controller 3
1.4. (dV) Reasonable damping ratio D for the closed control loop 4

2. Experiment T5: RetBot, distance control **5**
2.1. e) Simulink simulation of the closed control loop 5
2.2. f) Reference, manipulated and controlled variable 6
2.3. g) P-controller implementation on the LEGO robot 7
2.4. h) Video: robot approaching an obstacle 9
2.5. i) Recorded test-drive vs. simulation 10
2.6. j) Critical gain and rate-limit-induced oscillation 11

A. Appendix Frontmatter **13**
A.1. Declarations 13

B. Appendix Code and Resources **15**
B.1. Code used for the analysis of the Preperations 15

1 Preparations

1.1. (aV) P controller in the time domain

(aV) What is the equation that describes a digital P controller in the time domain? [1]

$$x_a(t) = K_P \cdot e(t) \tag{1.1}$$

1. $x_a(t)$: Input variable of the P controller
2. K_p: Proportional coefficient
3. $e(t)$: Control output of the P controller

1.2. (bV) Transfer function of open loop P controller

(bV) What is the transfer function of the open loop with P controller? Which characteristic shows $G_o(s)$? [1]

The transfer function $G_o(s)$ is obtained using the block diagram determined in the fourth experiment as follows:

$$G_o(s) = K_P \cdot \frac{-K_I}{T_1 \cdot s^2 + s} \tag{1.2}$$

$G_o(s)$ exhibits IT1 behavior.

1.3. (cV) Transfer function of distance control with P controller

(cV) What is the transfer function of the distance control with P controller? Which characteristic shows $G_W(s)$? [1]

By inserting 1.2, we obtain the following for $G_W(s)$:

$$G_W(s) = \frac{G_o(s)}{1 + G_o(s)} = \frac{1}{1 + \frac{-T_1}{K_P \cdot K_I} \cdot s^2 - \frac{1}{K_P \cdot K_I} \cdot s} \tag{1.3}$$

The closed-loop system $G_W(s)$ exhibits PT_2 behavior.

1.4. (dV) Reasonable damping ratio D for the closed control loop

(dV) Select a reasonable damping ratio D for the closed control loop and calculate the controller parameter K_P so that the desired/selected damping ratio is achieved. [1]

The damping ratio is set to D = 1 (aperiodic limit case) to ensure a fast, overshoot-free settling. This allows the robot to reach the target distance as quickly as possible without having to move backward. The standard mathematical form of a PT2 system is:

$$G_s = \frac{K_P}{\frac{1}{\omega_0^2} \cdot s^2 + \frac{2D}{\omega_0 \cdot s} + 1} \tag{1.4}$$

Rearranging 1.4 yields:

$$\frac{1}{\omega_0^2} = -\frac{T_1}{K_P \cdot K_I} \rightarrow \omega_0 = \sqrt{-\frac{K_P \cdot K_I}{T_1}} \tag{1.5}$$

$$\frac{2D}{\omega_0} = -\frac{1}{K_P \cdot K_I} \rightarrow \omega_0 = -K_P \cdot K_I \cdot 2 \cdot D \tag{1.6}$$

If we now set Equation 1.6 equal to 1.5, we can solve for K_P.

$$\sqrt{-\frac{K_P \cdot K_I}{T_1}} = -K_P \cdot K_I \cdot 2 \cdot D \rightarrow K_P = \frac{1}{4 \cdot D^2 \cdot T_1 \cdot K_I} \tag{1.7}$$

Using the previously determined parameters (mean value) K_I and T_1,
Left motor:

$$K_{I,L} = 0{,}437\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,L} = 57{,}1\,\mathrm{ms} = 0{,}0571\,\mathrm{s} \tag{1.8}$$

Right motor:

$$K_{I,R} = 0{,}446\,\mathrm{cm\,s^{-1}\,\%^{-1}}, \qquad T_{1,R} = 52{,}2\,\mathrm{ms} = 0{,}0522\,\mathrm{s} \tag{1.9}$$

the following values are obtained:
Left motor:

$$K_{P,L} = \frac{1}{4 \cdot 1^2 \cdot 0.0571 \cdot 0.437} = 9{,}99\,\%\,\mathrm{cm^{-1}} \approx 10\,\%\,\mathrm{cm^{-1}} \tag{1.10}$$

Right motor:

$$K_{P,R} = \frac{1}{4 \cdot 1^2 \cdot 0.0522 \cdot 0.446} = 10{,}77\,\%\,\mathrm{cm^{-1}} \approx 11\,\%\,\mathrm{cm^{-1}} \tag{1.11}$$

minus werte müssen rauskommen

2 Experiment T5: RetBot, distance control

2.1. e) Simulink simulation of the closed control loop

(e) Simulate the idealized behavior of the closed control loop in Matlab/Simulink using the controller parameter K_P calculated in the preparation by using the setup from previous experiment. [1]

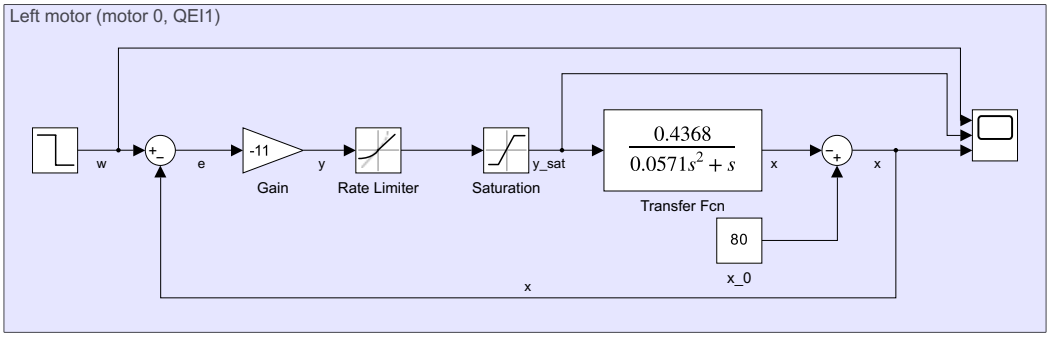


Abbildung 2.1.: Image of the simulink controller

cut of white space from simulink image print on pdf

Figure 2.1 shows the Simulink model used for the simulation of the closed control loop. The model is based on the one from the previous experiment, with the addition of a P controller block. The parameters for the simulation are set according to the values calculated in section 1.4.

2.2. f) Reference, manipulated and controlled variable

(f) Plot the reference variable, the manipulated variable, and the controlled variable via Matlab (clean graphs, no screenshots from the scope). Is the damping level you selected correct? Interpret the signal curves. [1]

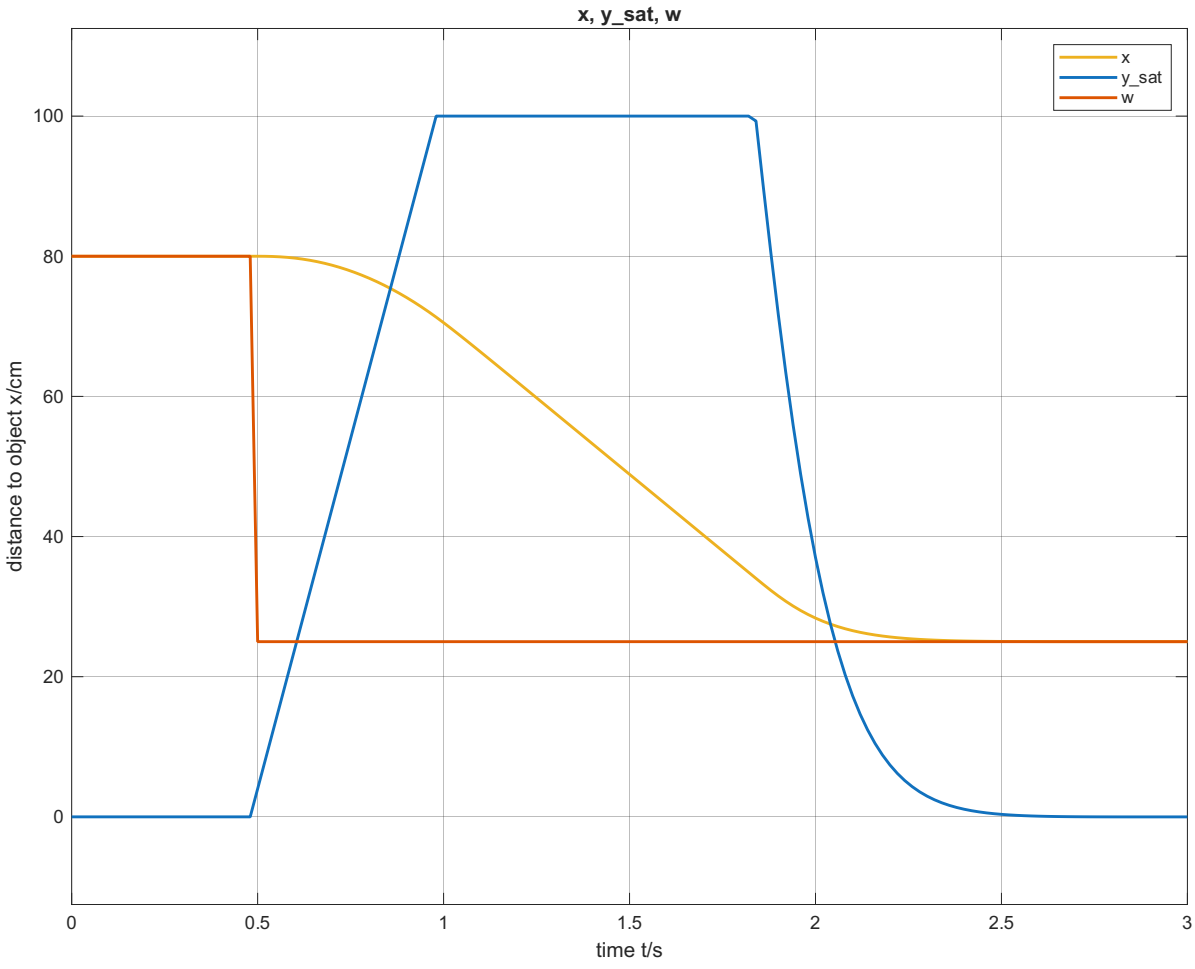


Abbildung 2.2.: Plot of the reference variable, the manipulated variable, and the controlled variable

interpret dampening level

- 1. Evaluation of the damping: The selected damping proves to be well-suited. Simulation and measurement show that the controlled variable x (actual distance)

The following section is a translation of the original Ger-

rapidly converges to the reference variable w (setpoint distance) without significant overshoot or instability. The system exhibits the desired aperiodic transient response, which is ideal for distance control to avoid collisions.

- 2. Interpretation of the Signal Curves:
  Reference variable w(t): It represents a step change from 80 cm to 25 cm.

  Manipulated variable y(t): At the start, the control error is at its maximum ($e = 80\,\text{cm} - 25\,\text{cm} = 55\,\text{cm}$). The P controller theoretically calculates an extremely high manipulated variable from this: $y = K_P \cdot e = 8{,}434 \cdot 55 \approx 464\,\%$. However, since the actual EV3 driver is limited to $\pm 100\,\%$, the motor is initially in saturation (full throttle"). The data shows that the control variable y does not drop significantly until shortly before reaching the target (in the range from approximately 1,04 s to 1,06 s. This marks the transition from saturation to the linear control region, in order to bring the robot to a stop at the target value.

  Controlled variable x(t): The distance initially decreases linearly, since the motor runs at a constant maximum speed. As soon as the manipulated variable y decreases, the curve flattens out, and the robot asymptotically approaches the target value of 25 cm. Thanks to the integrative behavior of the system (IT1 system), the final value is reached with stationary accuracy.

- 3. Necessary nonlinear element: To correctly model the real-world conditions in the simulation, a saturation element must be added. As calculated above, the linear P controller initially outputs values well above 100 %. The saturation element limits the control variable y in the model to the physically possible range $[-100, 100]$, which corresponds to the actual behavior of the motor driver.

use
qty for the units

2.3. g) P-controller implementation on the LEGO robot

(g) Implement the P controller for the LEGO robot - see 3.7 for the existing software-based control loop, that can be parameterized by using following commands using Serial.ICE:

- *'1': set* K_P
- *'2': set* K_I
- *'3': set* K_D
- *'4': set desired distance/cm*
- *'5': set* K_0
- *'6': set* K_1

Show the correct implementation by Serial.ICE commands and received data. [1]

explain what has been send

For the measurement, the following values were transmitted to the LEGO robot via Serial.ICE:

Listing 2.1: Serial.ICE commands send

```
r       -> QEI reset
m0      ->
1-11    -> set K_P to "-11"
20      -> set K_I to "0"
30      -> set K_D to "0"
425     -> set desired distance to "25" cm
50.99   -> set K_0/Motor0 to "0.99" rotations for each rotation
61      -> set K_1/Motor1 to "1" rotation for each rotation
```

The received data from the robot after sending the commands is shown in Listing 2.2.

Listing 2.2: Serial.ICE commands received

```
QEI-Reset done
Motor-state: 0
PID-Kp: -11.000000
PID-Ki: 0.000000
PID-Kd: 0.000000
Desired-dist/cm: 25.000000
K0 set to 0.990000
K1 set to 1.000000
fsample=500Hz
Time/ms;QEI1/steps;QEI2/steps;d/cm;Mot0/%
   0;    0;    0; 81.0;  3
   2;    0;    0; 81.0;  3
   4;    0;    0; 81.0;  3
   6;    0;    0; 81.0;  3
   8;    0;    0; 81.0;  3
  10;    0;    0; 81.0;  3
```

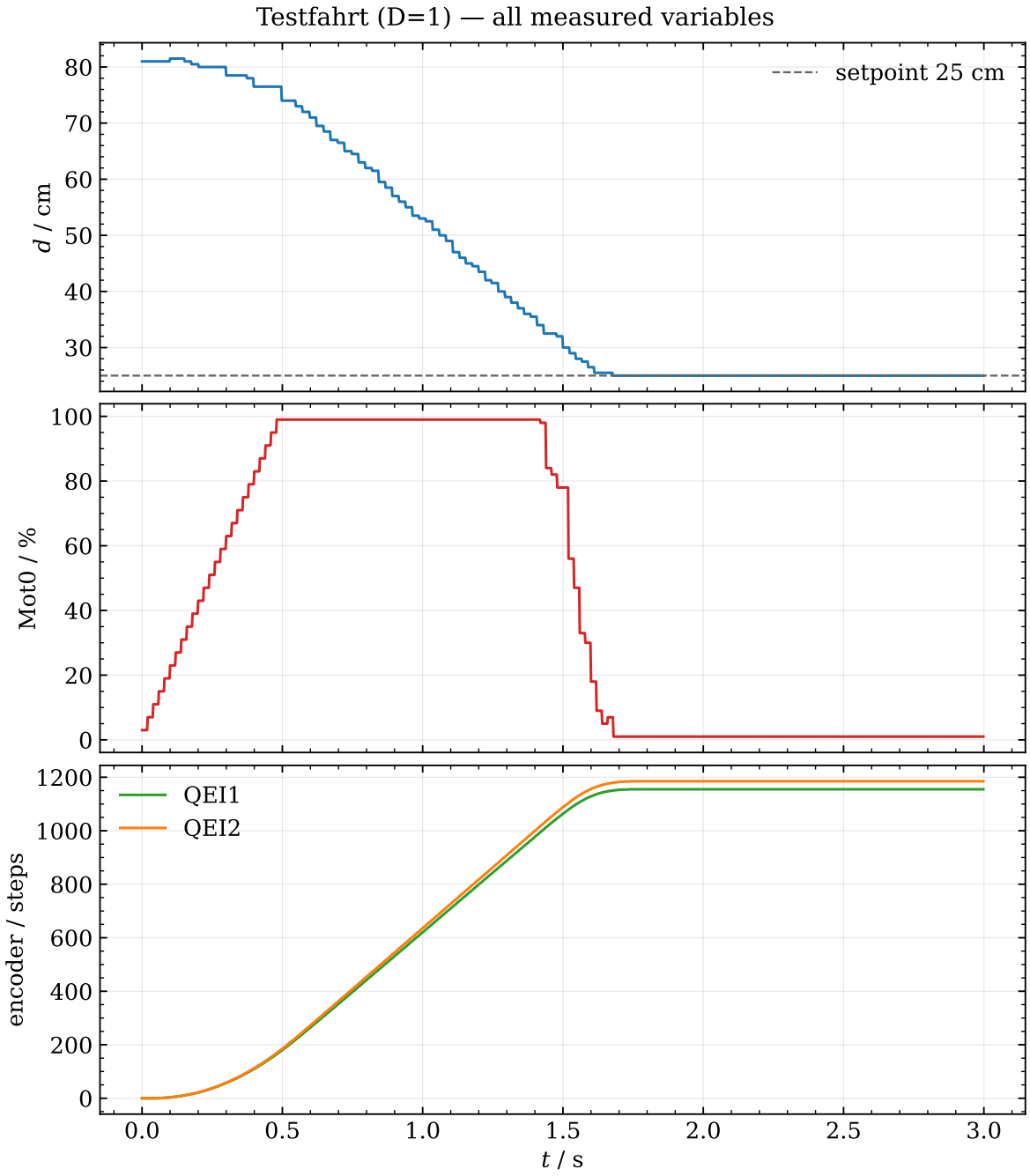
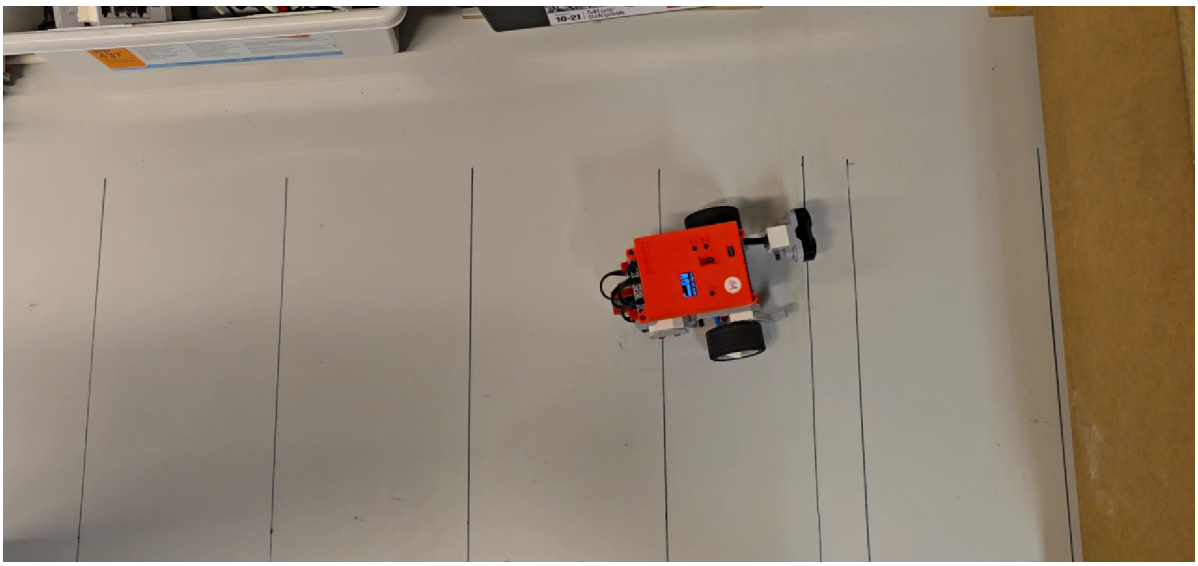


Abbildung 2.3.: QEI und Distance Retrun from Serial.ICE

add legend to all subplots of the graph or add detailed description in the caption

2.4. h) Video: robot approaching an obstacle

(h) Upload a video showing the robot moving towards an obstacle. [1]



https://youtu.be/gG7SXy00EJQ

2.5. i) Recorded test-drive vs. simulation

(i) Read the recorded data of your test-drive using Serial.ICE and compare it to the simulated results - distance, travelled distance and duty-cycle of motor 0! This has to be done in one graph after importing the data according to the previous experiment! [1]

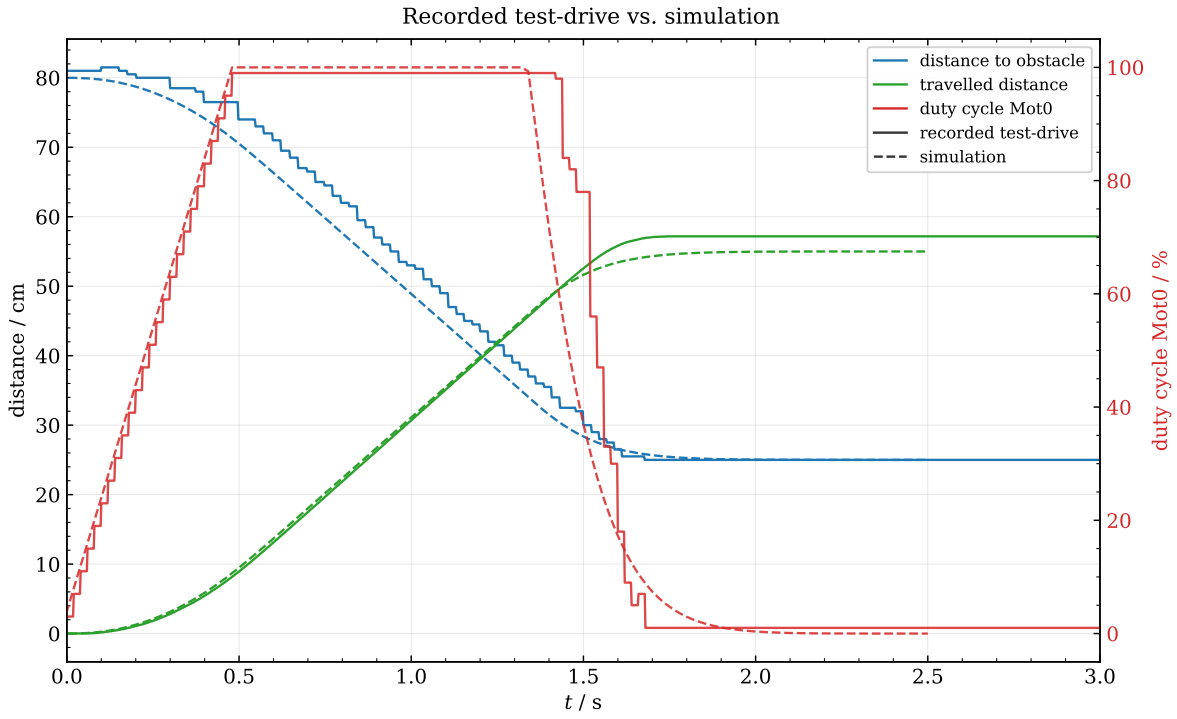


Abbildung 2.4.: Recorded test-drive vs. simulation: distance, travelled distance and duty cycle of motor 0.

compare test to simulation data + error analysis

Stepped curve (quantization): The measurement curve is not smooth but takes on a visible staircase shape. This is due to the digital resolution of the sensor, which does not output the distance continuously but only in discrete steps (e.g., in 1 mm increments).

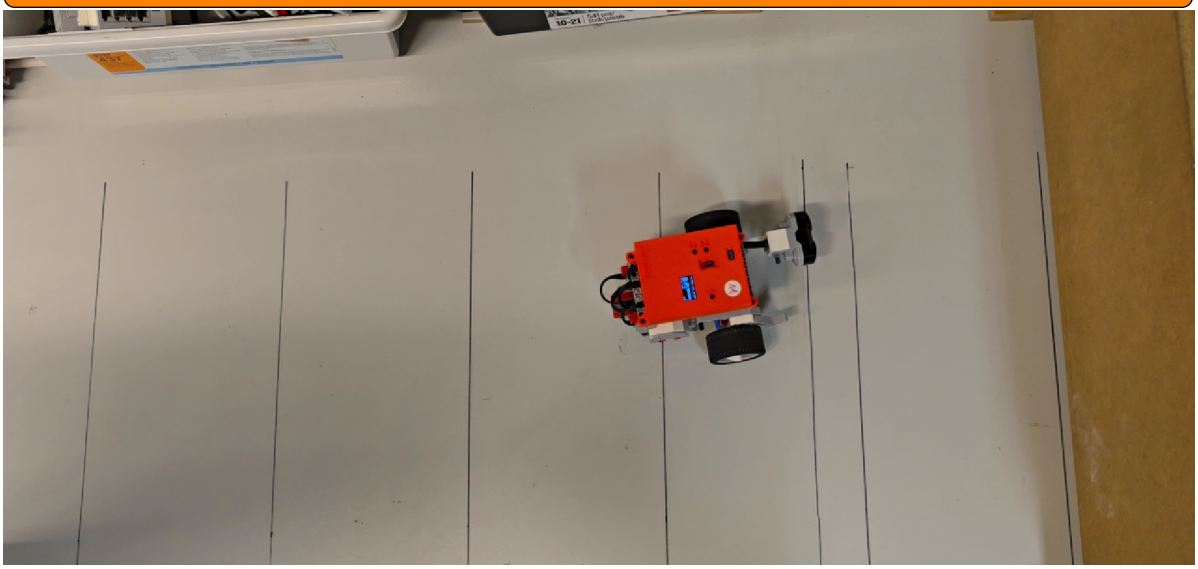
explain why the measured curve is not smooth but stepped in more detail and for the mesurment of the ultrasonic sensor and the motors

2.6. j) Critical gain and rate-limit-induced oscillation

> *(j) Increase K_P, check the influence of the rate-limit-factor in a way that the robot shows periodic oscillations - write down this critical value(s) and upload a video.* [1]

explain critical value(s) of D

explain dampening factor



`https://youtu.be/cy5t8bU4MxE`

$$K_{P,R} = \frac{-1}{4 \cdot 0.50^2 \cdot 0.0522 \cdot 0.446} = -42.95 \tag{2.1}$$

With $D = 0.5$ to achieve periodic oscillations, the critical value for K_P is $-42{,}95$
The video shows the robot oscillating around the target distance for this the following Serial.ICE commands were sent:

Listing 2.3: Serial.ICE commands and received datalog

```
r
m0
1-42
20
30
```

```
425
50.99
61
```

oscilation not perfekt, deadtime through sampleing rate.

rate limit factor is implemented for high speeds when robot is using full motor speed as this results in weel slipage and erros in all mesurments

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

”	Marius Off	Matrikelnummer: 3416002,
”	Janis Graf	Matrikelnummer: 3377251

Ort, Datum und Unterschrift

Abbildungsverzeichnis

2.1. Image of the simulink controller 5
2.2. Plot of the reference variable, the manipulated variable, and the controlled variable 6
2.3. QEI und Distance Retrun from Serial.ICE 9
2.4. Recorded test-drive vs. simulation: distance, travelled distance and duty cycle of motor 0. 10

B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time only about 1000 lines and a simulink model) is ***not*** included in this appendix. Including it as a verbatim listing still caused major build instability again again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T5-SoSe2026.git`

Literatur

[1] Torsten Reitmeier. *Internship in Control Engineering (RTP)*. Praktikumsversuche. 2025.